Data Structures and Algorithms Lab-Manual

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Implementation of a list (linear and binary search) using array ADT

Aim:

To implement a List using the Array Abstract Data Type (ADT) and perform Linear Search and Binary Search operations to search for an element in the list efficiently.

Pseudocode:

START

Read the size of the array (N).

Read N elements into the array.

Read the element to be searched (KEY).

// Linear Search

4. Set FOUND = FALSE.

5. For i = 0 to N - 1 do

If ARRAY[i] == KEY then

Print "Element found using Linear Search at position", i + 1.

FOUND = TRUE.

Break.

End If

End For

If FOUND == FALSE then

Print "Element not found using Linear Search."

End If

// Binary Search (Array must be sorted)

7. Sort the array in ascending order.

8. Set LOW = 0, HIGH = N - 1.

9. Set FOUND = FALSE.

10. While LOW <= HIGH do MID = (LOW + HIGH) / 2 If ARRAY[MID] == KEY then Print "Element found using Binary Search at position", MID + 1. FOUND = TRUE. Break. Else If KEY < ARRAY[MID] then HIGH = MID - 1 Else LOW = MID + 1 End If End While 11. If FOUND == FALSE then Print "Element not found using Binary Search." End If STOP

Code:

```
#include <stdio.h>


int main() {

   int a[5] = {9, 8, 6, 10, 7};

   int key, i, low = 0, high = 4, mid;

   int found = 0;


   printf("Enter element: ");

   scanf("%d", &key);


      for (i = 0; i < 5; i++) {

      if (a[i] == key) {

         printf("Linear Search: Found at position %d\n", i + 1);
```

```
        found = 1;
        break;
    }
}
if (found == 0)
    printf("Linear Search: Not Found\n");



found = 0;
while (low <= high) {
    mid = (low + high) / 2;

    if (a[mid] == key) {
        printf("Binary Search: Found at position %d\n", mid + 1);
        found = 1;
        break;
    } else if (key < a[mid]) {
        high = mid - 1;
    } else {
        low = mid + 1;
    }
}

if (found == 0)
    printf("Binary Search: Not Found\n");
```

```
    return 0;
}
```

Input and Output:

```
Output

Enter element: 8
Linear Search: Found at position 2
Binary Search: Not Found


=== Code Execution Successful ===
```

Result:

Therefore, the implementation of a List using the Array ADT has completed successfully, and the required element is searched using both Linear Search and Binary Search techniques. The search results are displayed .